# BCI-ROM Matrices Standalone Solver

For Simcenter™ Flotherm™ and Simcenter™ Flotherm™ XT

Software Version 2404

SIEMENS

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# Chapter 1
# BCI-ROM Matrices Standalone Solver

You can use the BCI-ROM Matrices Standalone Solver to solve BCI-ROM Matrices models created using Simcenter™ Flotherm™ 2019.1 or later or Simcenter™ Flotherm™ XT 2021.1 or later as either steady state or transient solutions.


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## Workflow

How to use the solver.

### Prerequisites

- You have exported a BCI-ROM Matrices model from Simcenter Flotherm or from Simcenter Flotherm XT.
- You have downloaded the solver from Support Center.

### Procedure

1. If your model folder is zipped, unzip it.
2. Create a directory to use as your solution directory and place the model folder in the solution directory.
3. Adjust BCI-ROM parameters as necessary. See “Modifying Input Parameters” on page 5.
4. (Optional) Create a transient time grid file, called *transient.txt*, in the solution directory. See -timeStep timestep.
5. (Optional) Create a source power definition file in the solution directory. See -power “power_definition_file”.
6. Run the solver command in a command prompt. See “Command Line Operation” on page 8.

### Results

The results are output to the solution directory.

## Modifying Input Parameters

You can specify BCI-ROM parameters to use during the solve by editing a copy of the BCI-ROM's *xData.txt* file.

## Procedure

1. Make a copy of the *xData.txt* found in the BCI-ROM model folder and place the copy in the location you will specify for -solutionPath when you run the solver command. The parameters in the copy in the -solutionPath location will overwrite the values read-in from the *xData.txt* in the unzipped BCI-ROM model.

2. Modify the following values as necessary:

**Table 1. BCI-ROM Input Parameters**

| Field | Description |
|---|---|
| SourceValues | This section lists the default power (in Watts) assigned to each source in the Simcenter Flotherm or Simcenter Flotherm XT model from which the BCI-ROM was created.<br>The order of the values in the SourceValues section reflects the order of the source names in the SourceNames section, that is, the first source value in the SourceValues section applies to the first source in the SourceNames list.<br>**Note:**<br>The values in this section will be overridden if you use the -power command line option. |
| HTCValues | This section lists the heat transfer coefficient values (in $W/m^2K$) defined in the Simcenter Flotherm or Simcenter Flotherm XT model from which the BCI-ROM was created.<br>The order of the values in the HTCValues section reflects the order of the HTC names in the HTCNames section, that is, the first HTC value in the HTCValues section applies to the first HTC in the HTCNames list.<br>**Note:**<br>Values entered must be within the HTC range specified when the BCI-ROM model was created. The range is given in the HTCRanges section of the *xData.txt* file. |
| HTCAmbients | This section lists the HTC ambient temperature values (in °C) specified in the Simcenter Flotherm or Simcenter Flotherm XT model from which the BCI-ROM was created.<br>Each value listed corresponds to an HTC value in the HTCValues section. Values are linked by the order they are listed, that is, the first value in the HTCAmbients section is linked to the first value in the HTCValue section, and they both correspond to the first name in the HTCNames list. |
| Tinitial | This section lists the initial temperature value (in °C) specified in the Simcenter Flotherm or Simcenter Flotherm XT model from which the BCI-ROM was created. |

## Examples

Example *xData.txt* file:

```
SourceNames:
BCI_Source1
BCI_Source2
ProbeNames:
BCI_MP1
```

```
BCI_MP2
BCI_Source1
BCI_Source2
HTCNames:
BCI_HTC1
BCI_HTC2
HTCRanges:
1.00000000e+00 1.00000000e+04
SourceValues:
2.00000000e+00
3.00000000e+00
HTCValues:
1.50000000e+01
4.50000000e+01
HTCAmbients:
3.50000000e+01
3.50000000e+01
Tinitial:
3.50000000e+01
ProbeSerials:
720AE32D12BF4BDB5F58861F0000482D
720AE32D12BF4BDB5F58861F0000482E
```

**Note:**
BCI-ROM extraction automatically generates probes collocated with the defined sources.

# Command Line Operation

The standalone solver runs from the command line.

## Usage

BCIMatricesSolver.exe **–bciPath** “***input_path***” **–solutionPath** “***solution_path***” [options]

## Arguments

- **-bciPath** “***input_path***”

  Specifies the input directory. This is the folder that contains the **.mtx* BCI-ROM Matrices created by Simcenter Flotherm or Simcenter Flotherm XT. Files contained here should not be modified directly.

- **-solutionPath** “***solution_path***”

  Specifies the solution directory. This is the folder where all the results of the specified solve and user-defined input parameters will be placed.

- -steadyState

  Specifies the type of solve to be steady state. The *absence* of this option will automatically specify the type of solve to be transient.

- -timeStep *timestep*

  Specifies the step size, in seconds, of a transient solve. This will apply a uniform transient time grid with the specified step size.

  A transient profile defined in a **.txt* file can also be used. The file needs to be named *transient.txt* and placed in the defined solution directory, and then the transient profile will automatically be used. An example of the contents of a *transient.txt* file is:

  ```
  0.0
  1.0
  2
  5
  10
  15
  20
  ```

  The transient steps can be non-linear (as in the above example). This is the time from start to finish of the solve.

- -endTime *time*

  Specifies the end time, in seconds, for a transient solve.

- -saveRomState

  Specifies saving the end state of a solve in a format that can be used to initialize a future solve. The file will be placed in the *same directory as the defined solutionPath*. The file will either be named “endRomStateS” or “endRomStateT” for a steady state or transient solve, respectively.

- -loadRomState “*path_to_saved_state*”

Specifies using the saved ROM state as the initial condition for the solve. The full path and file name with the extension are required.

- -power "*power_definition_file*"

  Specifies the use of a user-generated power input. The supported power definitions are: Constant, Pulse, Sinusoidal, PiecewiseLinear, and PID. All of the definitions are in Watts and the sub-functions will follow the same behavior as Simcenter Flotherm or Simcenter Flotherm XT transient attributes.

  The power definition is required to be a **.txt* file placed in the defined solutionPath. The contents must match the format specified in "Power Definition File Format" on page 11.

- -output *n*

  Specifies for a transient solve the desired sample points for results.

  - If -output is omitted, results for all time steps are output.
  - If -output is specified and *n* is omitted, only results for the last time step are output.
  - If -output is specified and *n* (an integer) is specified, the results of the first step, then the results of every $n^{th}$ step, and the results of the last step, are output.

## Examples

1. Steady-state solve, saving the end state:

   ```
   BCIMatricesSolver.exe -bciPath "c:\myBCIROM" -solutionPath "c:\BCI_Out
   put" -steadyState -saveRomState
   ```

   This will solve the BCI-ROM matrices found in *c:\myBCIROM* and output results to *c:\BCI_Output* in files called *ResultForProbes.txt*, and *ResultForSurfaces.txt*. Examples of the content of these files is:

   - ResultForProbes.txt:

     ```
     Probe temperatures:
     Probe [BCI_MP1] = 1.2831e+02
     Probe [BCI_MP2] = 1.1906e+02
     Probe [BCI_Source1] = 1.1617e+02
     Probe [BCI_Source2] = 1.1360e+02
     ```

   - ResultForSurfaces.txt:

     ```
     Face temperatures and Fluxes:
     Face [BCI_HTC1] T Flux = 5.3116e+01 -2.0000e+01
     ```

2. Transient solve, loading saved state, and with user defined power input:

   ```
   BCIMatricesSolver.exe -bciPath "c:\myBCIROM" -solutionPath "c:\BCI_Out
   put" -timeStep 0.1 -endTime 50 -loadRomState "c:\BCI_Output\endRomStat
   eT.txt" -power "BCI_Power.txt"
   ```

This will solve the BCI-ROM matrices found in *c:\myBCIROM* with a time step size of 0.1 seconds for the duration of 50 seconds, using a previous solve as the initial starting point, while also using a user-defined source behavior. The results of the solve will be outputted to *c:\BCI_Output* in files called *ResultForProbes.txt*, *ResultForSources.txt* and *ResultForSurfaces.txt*. Examples of the content of these files is:

- ResultForProbes.txt:

```
Transient probe temperatures: [BCI_MP1] [BCI_MP2] [BCI_Source1]
 [BCI_Source2]
Time=1.000000e-01 Probe T = 3.563427e+01 3.517015e+01 3.563552e+01
 3.658627e+01
Time=2.000000e-01 Probe T = 3.595375e+01 3.544953e+01 3.595476e+01
 3.738533e+01
Time=3.000000e-01 Probe T = 3.619200e+01 3.582710e+01 3.619109e+01
 3.798037e+01
Time=4.000000e-01 Probe T = 3.639382e+01 3.629968e+01 3.639057e+01
 3.848345e+01
```

- ResultForSources.txt:

```
Transient source values: [BCI_Source1] [BCI_Source2]
Time=1.000000e-01 Source values =  2.000000e+00 3.000000e+00
Time=2.000000e-01 Source values =  2.000000e+00 3.000000e+00
Time=3.000000e-01 Source values =  2.000000e+00 3.000000e+00
Time=4.000000e-01 Source values =  2.000000e+00 3.000000e+00
```

- ResultForSurfaces.txt:

```
Transient face temperatures and Fluxes: [BCI_HTC1]
Time=1.000000e-01 Face T Flux =  3.500106e+01 1.170741e-03
Time=2.000000e-01 Face T Flux =  3.500238e+01 2.632724e-03
Time=3.000000e-01 Face T Flux =  3.500394e+01 4.354048e-03
Time=4.000000e-01 Face T Flux =  3.500566e+01 6.244189e-03
```

# Power Definition File Format

Format of the power definition file.

Specify the filename using the -power command line option.

## Format

A **.txt* file placed in the defined solutionPath.

## Parameters

**Table 2. Power Definition File Commands**

| Command | Description |
|---|---|
| index <*indices*> | Indicates which sources the function is defined for. This can be for a single source or multiple sources. The source indices always start from 0. For example, the sources in a model with three sources will have 0, 1, and 2 as the value of their indices. |
| type {add\|multiply} | If multiple sub-functions are defined for source or set of sources, whether to combine the defined profiles by adding or multiplying. |
| periodic {true\|false} | Whether the function should repeat periodically until the end of the transient solve. |
| FunctionList | Indicates the beginning of the sub-function definition. This section can contain multiple sub-functions. |
| FunctionListEnd | Indicates the end of the defined sub-functions. |

The format for each power sub-function type is given in Table 3: Power Definition Sub-Functions.

**Table 3. Power Definition Sub-Functions**

| Sub-function Type | Format |
|---|---|
| Constant | FunctionType Constant<br>value <*value*> |
| Pulse | FunctionType Pulse<br>tStart <*value*><br>tEnd <*value*><br>amplitude <*value*><br>riseTime <*value*><br>highTime <*value*><br>fallTime <*value*> |

**Table 3. Power Definition Sub-Functions (continued)**

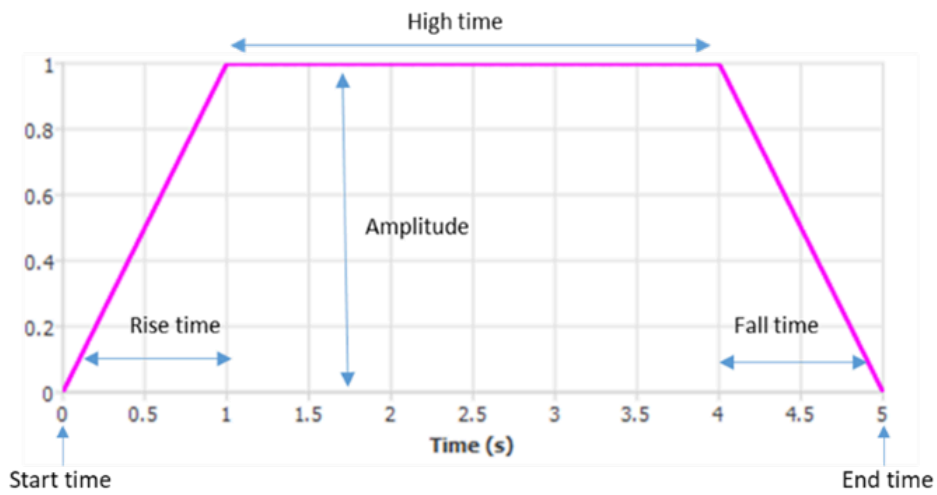
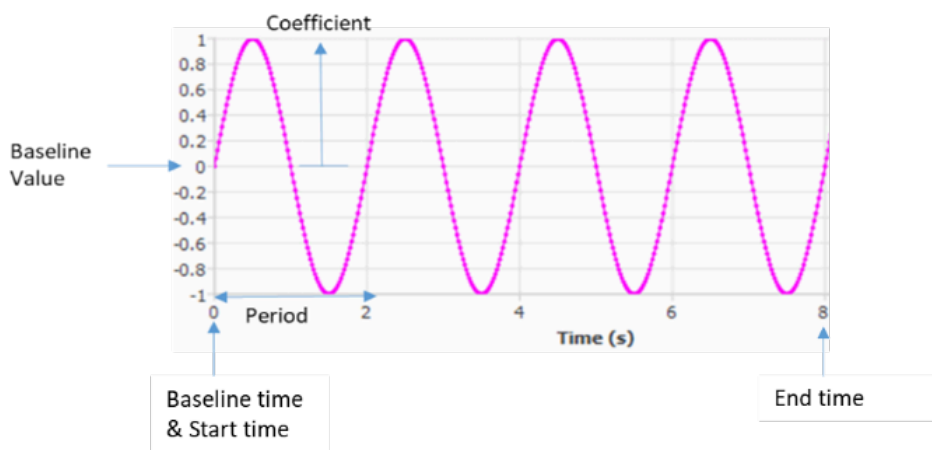
| Sub-function Type | Format | |
|---|---|---|
| |  | |
| Sinusoidal | FunctionType Sinusoidal<br>tStart <*value*><br>tEnd <*value*><br>baseLineValue <*value*><br>baseLineTime <*value*><br>coefficient <*value*><br>period <*value*><br>  | |
| PiecewiseLinear | FunctionType PiecewiseLinear<br>*<time_value> <power_value>*<br>*<time_value> <power_value>*<br>*<time_value> <power_value>* | |
| PID | FunctionType PID | *Description* |

**Table 3. Power Definition Sub-Functions (continued)**

| Sub-function Type | Format | |
|---|---|---|
| | targetValue <*value*> | Target temperature (°C) |
| | kp <*value*> | P coefficient |
| | ki <*value*> | I coefficient |
| | kd <*value*> | D coefficient |
| | maxPower <*value*> | Maximum allowable power (W) |
| | probeIndex <*value*> | Probe used for temperature monitoring |

**Note:**
The PiecewiseLinear and PID power definition cannot be mixed with other types of power definition within the same FunctionList for a source or sources. For example if source “A” has a PiecewiseLinear power sub-function defined, source “A” cannot have another type of power sub-function defined and combined with the PiecewiseLinear sub-function.

If any source is not defined within the power input file, the constant value specified in the input parameter file from Simcenter Flotherm or Simcenter Flotherm XTwill be used.

## Examples

Example of a power input file for a 3 source model, with explanatory comments in parentheses.

```
SourceDefinition:
index 0 2                     (Applied to sources with index 0 & 2)
type add
periodic false
FunctionList
FunctionType Constant
value 2
FunctionListEnd               (Ends the sub-function definition)
index 1
type add
periodic true                (If set to false the power becomes 0 after the
 end time)
FunctionList
FunctionType Pulse           (Specifies the sub-function applied to source
 with index 1)
tStart 0
tEnd 5
amplitude 5
riseTime 1
highTime 3
fallTime 2
FunctionType Sinusoidal         (The second sub-function applied to source
 with index 1)
tStart 0
tEnd 10
```

```
baseLineValue 3
baseLineTime 1
coefficient 3
period 4
FunctionListEnd
```